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ansys-magnet-segmentation-toolkit



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CONTENTS

The Magnet Segmentation Toolkit is a Python wrapper for automating the segmentation and skew of inner rotor interior permanent magnet (IPM) and surface permanent magnet (SPM) motors using Ansys Electronics Desktop (AEDT). You can launch this toolkit from the AEDT UI or launch it directly from a Python console.

Getting started Learn more about the Magnet Segmentation Toolkit and how to install it.

Getting started **UI reference** Understand how to use the Magnet Segmentation Toolkit Wizard.

UI reference **API reference** Understand the APIs available for the Magnet Segmentation Toolkit.

API reference **Examples** Explore example that shows how to use the Magnet Segmentation Toolkit to perform many different types of operations.

Examples **Contribute** Learn how to contribute to the Magnet Segmentation Toolkit codebase or documentation.

Contribute

GETTING STARTED

This section explains how to install the AEDT Magnet Segmentation Toolkit.

Installation Learn how to install the AEDT Magnet Segmentation Toolkit.

Installation **User guide** Learn more about the Magnet Segmentation wizard and how to use it.

User guide

1.1 Installation

1.1.1 Download links

The following installers are available for different operating systems:

Table 1: Available Installers

Installer Link	Operating System
Download	Windows
Download	Ubuntu 22.04
Download	Ubuntu 24.04

Visit the [Releases](#) page and pull down the latest installer.

1.1.2 Installing the Magnet Segmentation Toolkit

Windows

First step is installing the Magnet Segmentation Toolkit. In order to do so, follow the next steps.

1. Download the necessary installer from the [latest available release](#). The file should be named Magnet-Segmentation-Toolkit-Installer.exe.
2. Execute the installer.
3. Search for the Magnet Segmentation Toolkit and run it.

The Magnet Segmentation Toolkit window should appear at this stage.

Linux

Ubuntu

Prerequisites:

1. OS supported for **Ubuntu(24.04 and 22.04)**.

2. Update apt-get repository and install the following packages with **sudo** privileges: **wget, gnome, libffi-dev, libssl-dev, libsqlite3-dev, libxcb-xinerama0** and **build-essential** packages with **sudo** privileges

```
sudo apt-get update -y
sudo apt-get install wget gnome libffi-dev libssl-dev libsqlite3-dev libxcb-
↳xinerama0 build-essential -y
```

3. Install **zlib** package

```
wget https://zlib.net/current/zlib.tar.gz
tar xvzf zlib.tar.gz
cd zlib-*
make clean
./configure
make
sudo make install
```

To install the Magnet Segmentation Toolkit, follow below steps.

1. Download the necessary installer from the [latest available release](#). The file should be named Magnet-Segmentation-Toolkit-Installer-ubuntu_*.zip.
2. Execute the below command on the terminal

```
unzip Magnet-Segmentation-Toolkit-Installer-ubuntu_*.zip
./installer.sh
```

3. Search for the Magnet Segmentation Toolkit and run it.

The Magnet Segmentation Toolkit window should appear at this stage.

To uninstall the Magnet Segmentation Toolkit, follow below steps.

1. Go to File menu. Click Uninstall option.
2. Click Uninstall button.

1.1.3 Python Installation

The Magnet Segmentation Toolkit can be installed like any other open source package. From PyPI, you can either install both the backend and user interface (UI) methods or install only the backend methods.

You can either install both the backend and user interface (UI) methods or install only the backend methods.

To install both the backend and UI methods, run this command:

```
pip install ansys-magnet-segmentation-toolkit[all]
```

If you only need the common API, install only the backend methods with this command:

```
pip install ansys-magnet-segmentation-toolkit
```

1.1.4 For developers

You can be up and running with four lines of code:

```
git clone https://github.com/ansys/magnet-segmentation-toolkit
cd magnet-segmentation-toolkit
pip install -e .
```

Now you can run it with:

```
run_toolkit
```

Details

Installing Pytools installer in developer mode allows you to modify the source and enhance it.

Before contributing to the project, please refer to the [PyAnsys Developers guide](#). You need to follow these steps:

1. Start by cloning this repository:

```
git clone https://github.com/ansys/magnet-segmentation-toolkit
```

2. Create a fresh-clean Python environment and activate it. Refer to the official [venv](#) documentation if you require further information:

```
# Create a virtual environment
python -m venv .venv
# Activate it in a POSIX system
source .venv/bin/activate
# Activate it in Windows CMD environment
.venv\Scripts\activate.bat
# Activate it in Windows Powershell
.venv\Scripts\Activate.ps1
```

3. Install the project in editable mode:

```
python -m pip install -e .[tests,doc]
```

4. Finally, verify your development installation by running:

```
pytest tests -v
```

Style and testing

This project uses [pre-commit](#). Install with:

```
pip install pre-commit
run pre-commit install
```

This now runs [pre-commit](#) for each commit to ensure you follow project style guidelines. For example:

```
git commit -am 'fix style'
isort.....Passed
black.....Passed
blacken-docs.....Passed
flake8.....Passed
codespell.....Passed
pydocstyle.....Passed
check for merge conflicts.....Passed
debug statements (python).....Passed
check yaml.....Passed
trim trailing whitespace.....Passed
Validate GitHub Workflows.....Passed
```

If you need to run it again on all files and not just staged files, run:

```
run pre-commit run --all-files
```

Local build

This app can be deployed as a frozen app using `pyinstaller` with:

```
pip install -e .[freeze]
run pyinstaller frozen.spec
```

This generates app files at `dist/magnet_segmentation_toolkit` and you can run it locally by executing `Magnet Segmentation Toolkit.exe`. For more information on how to create a standalone executable, refer to this section: [How to distribute](#).

Documentation

For building documentation, you can either run the usual rules provided in the [Sphinx](#) Makefile:

```
pip install -e .[doc]
doc/make.bat html
# subsequently open the documentation with (under Linux):
<your_browser_name> doc/html/index.html
```

1.2 User guide

You have multiple options for installing and launching the Magnet Segmentation Toolkit:

- You can install the toolkit directly in AEDT using an installation script and then launch it as a wizard. For more information, see *Install toolkit in AEDT and launch the Magnet Segmentation Toolkit*.
- You can install the toolkit from a Python console and then launch the Magnet Segmentation Toolkit. For more information, see *Install toolkit from Python console and launch the Magnet Segmentation Toolkit*.
- You can install the toolkit from a Python console and then use the toolkits APIs. For more information, see *Install toolkit from Python console and use the toolkits APIs*.

1.2.1 Install toolkit in AEDT and launch the Magnet Segmentation Toolkit

You can install the Magnet Segmentation Toolkit directly in AEDT using the base interpreter from the AEDT installation.

1. From [Install from a Python file](#), follow the steps to install PyAEDT inside AEDT.
2. In AEDT, select **Tools > Toolkit > PyAEDT > Console** to load the PyAEDT console:

```

Python 3.7.13 (remotes/origin/3b89b4a151d5e27a7d119919e370e421549562b8-dirty:3b89b4a1, Sep 23 2) [MSC v.1920 64 bit (AMD64)]
Type 'copyright', 'credits' or 'license' for more information
IPython 7.34.0 -- An enhanced Interactive Python. Type '?' for help.
Loading the PyAEDT Console.
pyaedt INFO: using existing logger.
pyaedt INFO: Launching PyAEDT outside AEDT with CPython and PythonNET.
pyaedt INFO: AEDT installation Path C:\Program Files\AnsysEM\v231\Win64.
pyaedt INFO: Launching AEDT with module PythonNET.
pyaedt INFO: AEDT 2023.1 Started with process ID 16440.
pyaedt INFO: pyaedt v0.6.71
pyaedt INFO: Python version 3.7.13 (remotes/origin/3b89b4a151d5e27a7d119919e370e421549562b8-dirty:3b89b4a1, Sep 23 2) [MSC v.1920 64 bit (AMD64)]

*****
* ElectronicsDesktop 2023.1 Process ID 16440
* CPython 3.7.13
*****
* Example: hfss = pyaedt.Hfss()
* Example: m2d = pyaedt.Maxwell2d()
* Type exit() to close the console and release the desktop.
* desktop object is initialized and available. Example:
* desktop.logger.info('Hello world')
*****

In [1]:

```

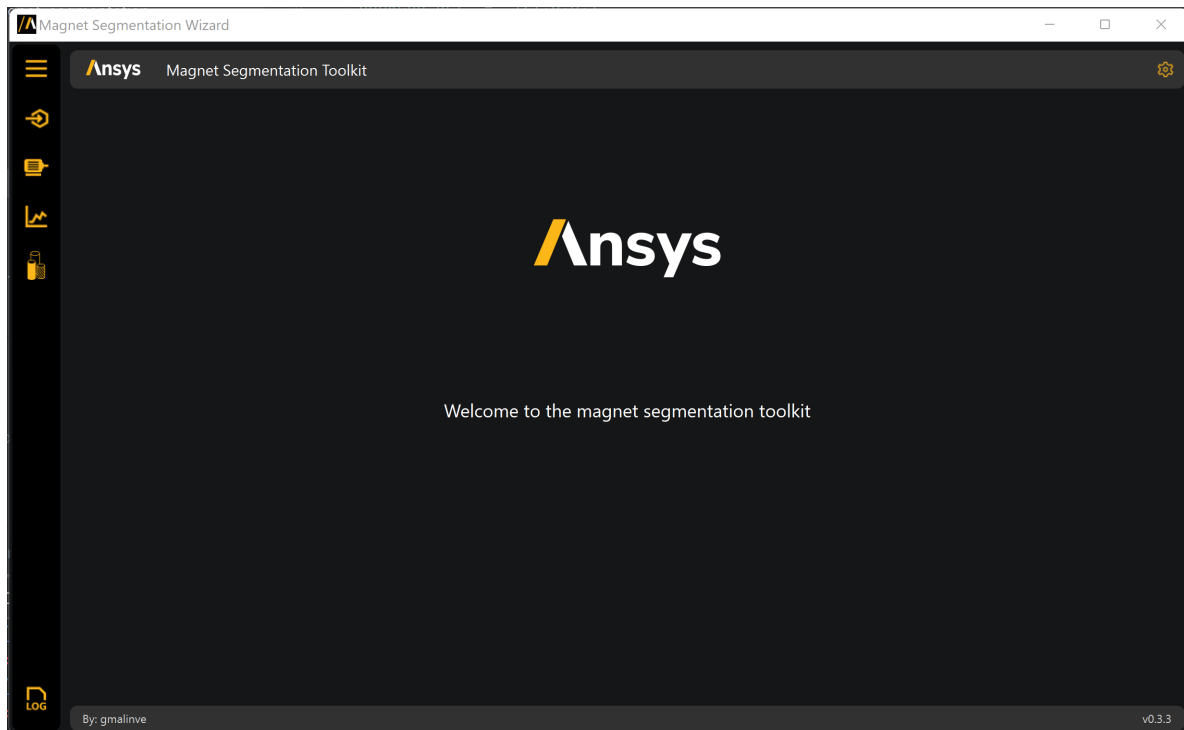
3. In the PyAEDT console, run these commands to add the Magnet Segmentation Toolkit as a wizard (toolkit UI) in AEDT:

```

desktop.add_custom_toolkit("MagnetSegmentationWizard")
exit()

```

4. In the AEDT toolbar, click the **MagnetSegmentationWizard** button to open this wizard in AEDT:



The wizard is connected directly to the AEDT session. For wizard usage information, see *UI reference*.

1.2.2 Install toolkit from Python console and launch the Magnet Segmentation Toolkit

You can install the Magnet Segmentation Toolkit in a specific Python environment from the AEDT console.

Note

If you have an existing virtual environment, skip step 1.

Note

If you have already installed the toolkit in your virtual environment, skip step 2.

1. Create a fresh-clean Python environment and activate it:

```
# Create a virtual environment
python -m venv .venv

# Activate it in a POSIX system
source .venv/bin/activate

# Activate it in a Windows CMD environment
.venv\Scripts\activate.bat

# Activate it in Windows PowerShell
.venv\Scripts\Activate.ps1
```

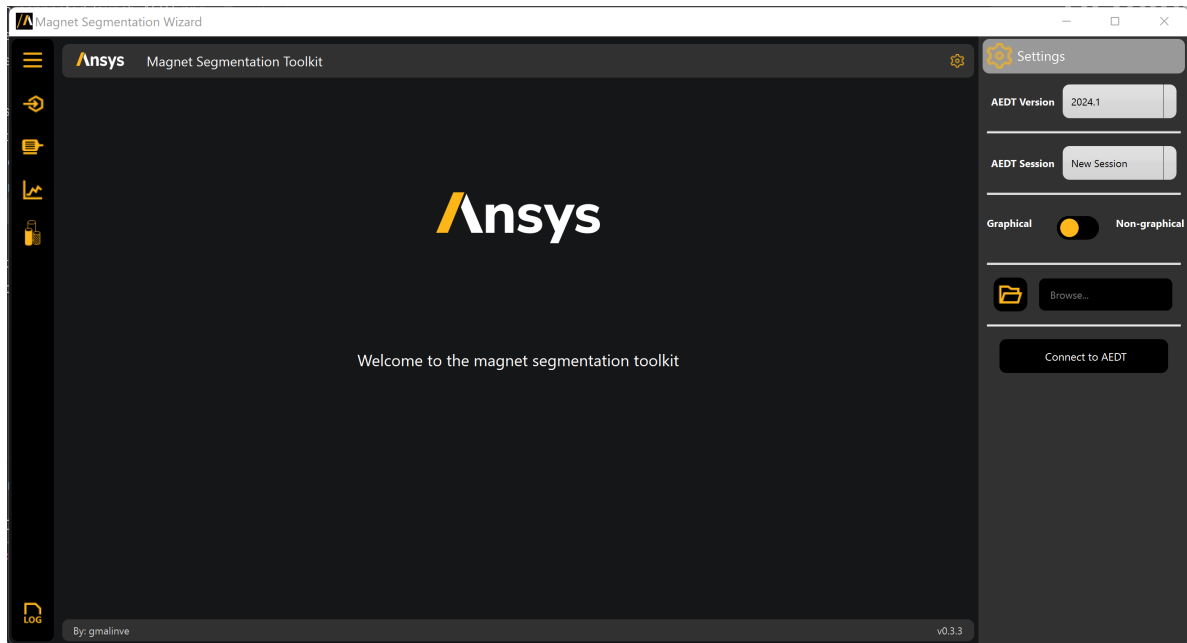
2. Install the toolkit from the GitHub repository:

```
python -m pip install ansys-magnet-segmentation-toolkit
```

3. Launch the Magnet Segmentation Toolkit Wizard:

```
python .venv\Lib\site-packages\ansys\aedt\toolkits\magnet_segmentation\run_toolkit.  
↪py
```

4. On the **AEDT Settings** tab, create an AEDT session or connect to an existing one:



For wizard usage information, see *UI reference*.

1.2.3 Install toolkit from Python console and use the toolkits APIs

You can install the toolkit in a specific Python environment and use the toolkits APIs. The code example included in this topic shows how to use the APIs at the model level and toolkit level.

Note

If you have an existing virtual environment, skip step 1.

Note

If you have already installed the toolkit in your virtual environment, skip step 2.

1. Create a fresh-clean Python environment and activate it:

```
# Create a virtual environment
python -m venv .venv

# Activate it in a POSIX system
source .venv/bin/activate

# Activate it in a Windows CMD environment
.venv\Scripts\activate.bat

# Activate it in Windows PowerShell
.venv\Scripts\Activate.ps1
```

2. Install the toolkit from the GitHub repository:

```
python -m pip install ansys-magnet-segmentation-toolkit
```

3. Open a Python console in your virtual environment:

```
python
```

4. From the command line, use the toolkit to perform segmentation and skew.

Use the toolkits APIs to import the toolkit, launch AEDT, open a 3D motor model, and then segment and skew this model in Maxwell 3D:

```
# Import backend services
from ansys.aedt.toolkits.magnet_segmentation.backend.api import _
↳ ToolkitBackend

# Backend object
toolkit = ToolkitBackend()

# Get service properties
properties = toolkit.get_properties()

# Define properties

properties["active_project"] = active_project
properties["active_design"] = active_design
properties["design_list"] = {active_project: [active_design]}
properties["is_skewed"] = False
properties["rotor_material"] = "M250-35A_20C"
properties["stator_material"] = "M250-35A_20C"
properties["magnets_material"] = "N30UH_65C"
properties["magnet_segments_per_slice"] = 2
properties["rotor_slices"] = 2
properties["apply_mesh_sheets"] = True
properties["mesh_sheets_number"] = 3

# Set service properties
toolkit.set_properties(properties)

# Launch AEDT, open project and connect to Maxwell3d design
toolkit.launch_aedt()
toolkit.open_project(aedt_file)
toolkit.connect_design("Maxwell3D")

# Segment and skew motor
toolkit.segmentation()
toolkit.apply_skew()

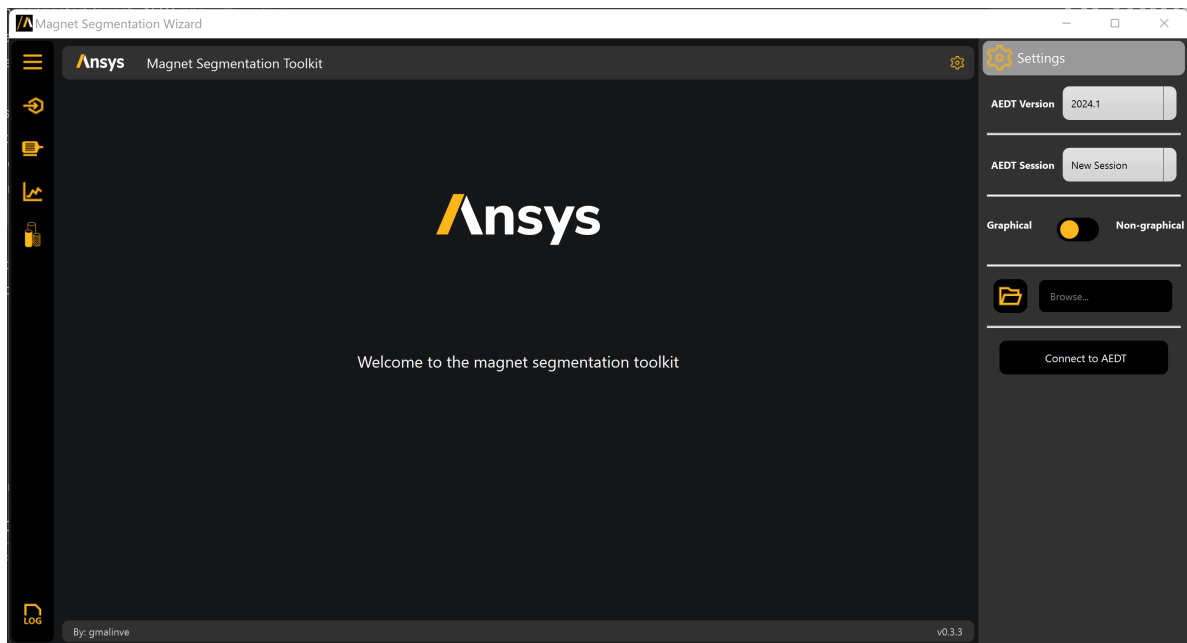
# Release AEDT
service.release_aedt()
```

For descriptions of the APIs available for the Magnet Segmentation Toolkit, see *API reference*.

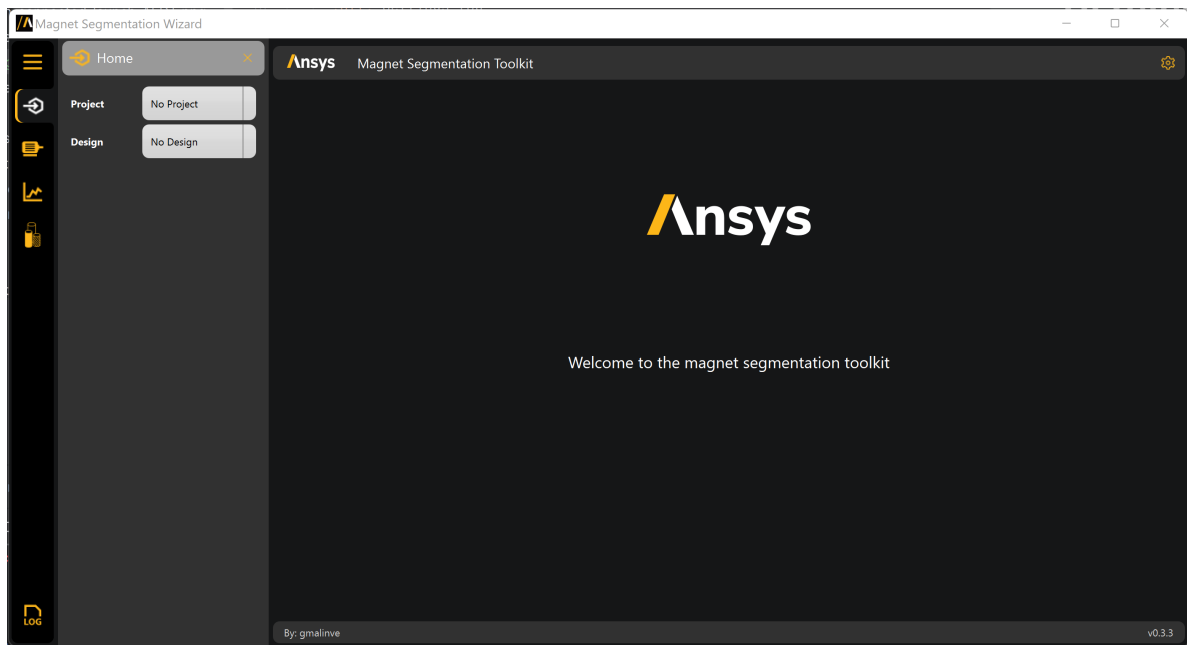
UI REFERENCE

This section describes how to use the Magnet Segmentation Toolkit Wizard, which requires an installed and licensed copy of AEDT. It assumes that you have already launched the wizard from either the AEDT menu or AEDT console. For toolkit installation and wizard launching information, see these topics:

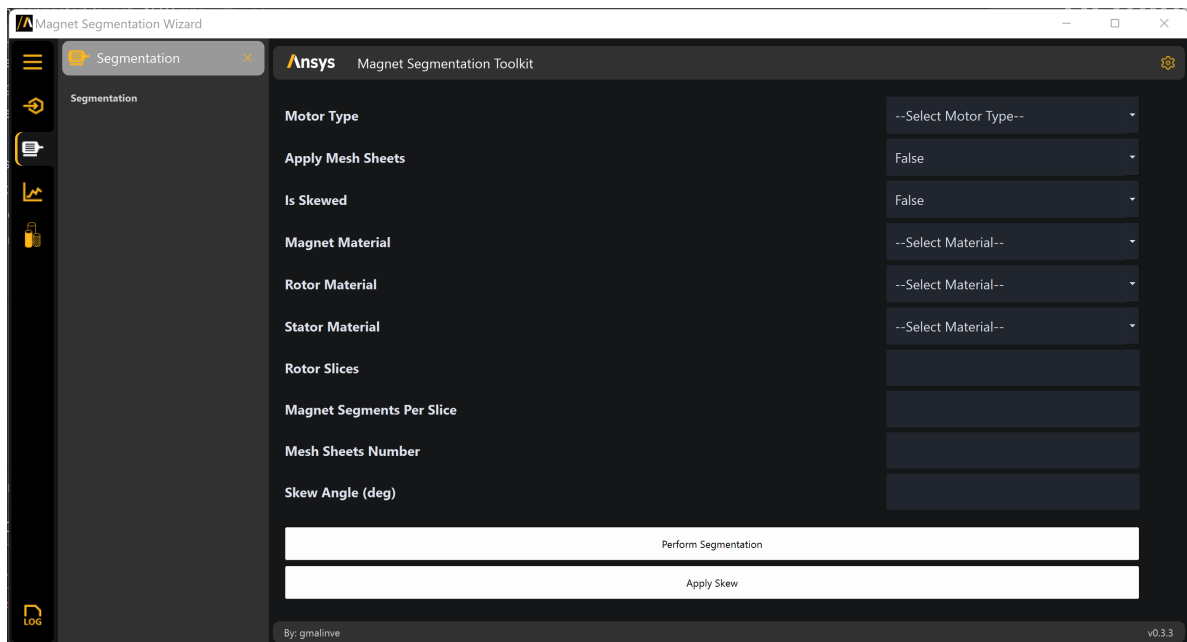
- *Install toolkit in AEDT and launch the Magnet Segmentation Toolkit*
 - *Install toolkit from Python console and launch the Magnet Segmentation Toolkit*
1. On the **Settings** tab, specify settings for either creating an AEDT session or connecting to an existing AEDT session and click **Connect to AEDT**.



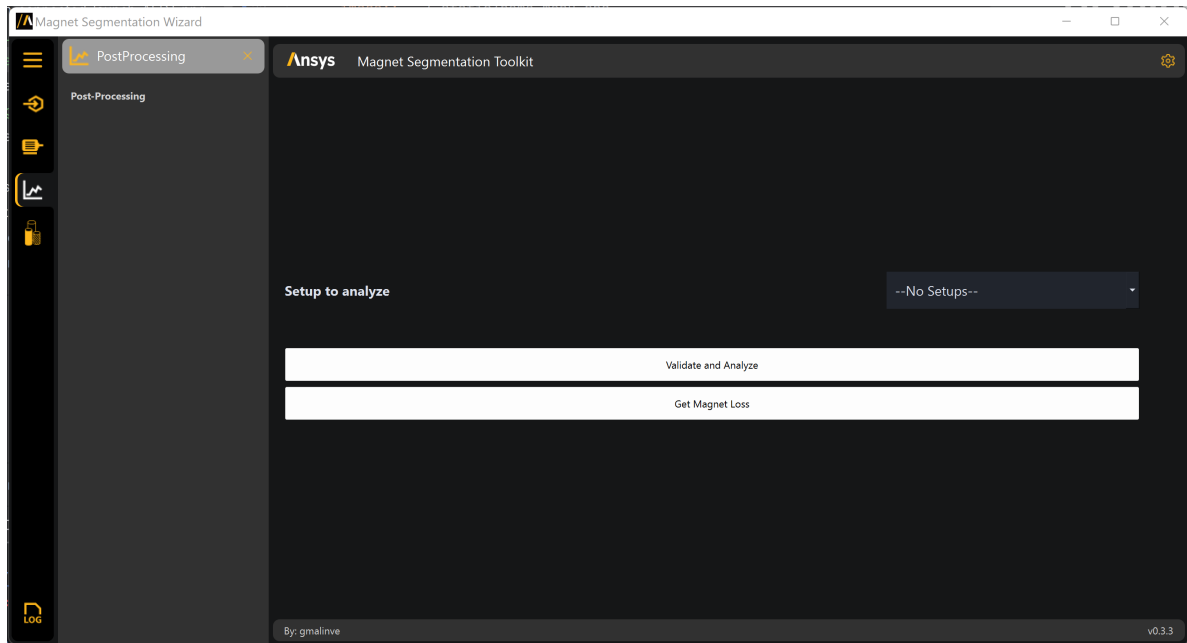
2. Choose project and design from the drop-down list in the **Home menu**.



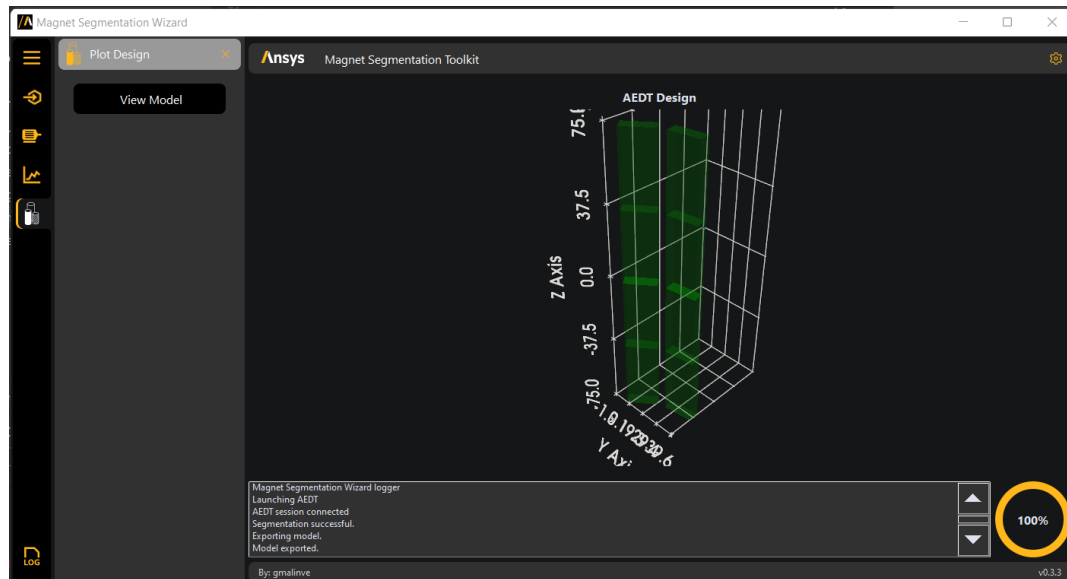
3. Click on the **Segmentation** menu to specify segmentation settings.



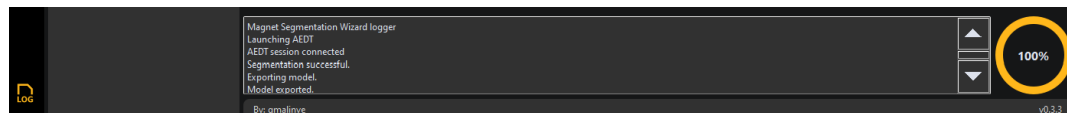
4. At the bottom of the tab, click **Perform Segmentation** and then **Apply Skew**.
5. Click on the **Post-processing** menu to select the desired setup to validate and analyze.



6. Click on **Get Magnet Loss** to automatically compute Magnet Loss in AEDT. The report is automatically generated in AEDT.
7. Click on the **Design** menu to visualize within the toolkit the segmented magnets.



8. The wizard has a progress circle and a logger box where you can see the status of every operation. Every operation must wait for the previous operation to release the toolkit.



API REFERENCE

This section provides descriptions of the available API for the Magnet Segmentation Toolkit:

- **Toolkit API:** Contains the `ToolkitBackend` class, which provides methods for controlling the toolkit workflow. In addition to methods for creating an AEDT session or connecting to an existing AEDT session, this API provides methods for automating the segmentation and skew of a 3D motor.

Warning

Both segmentation and skew of a 3D motor have requirements on the AEDT active project. Ensure that the active design meets these requirements:

- For segmentation, `SymmetryFactor` and `HalfAxial` design settings must be defined.
- For skew, `Shaft` must be the name of the shaft.

3.1 Toolkit API

The Toolkit API contains the `ToolkitBackend` class, which provides methods for controlling the toolkit workflow. In addition to methods for creating an AEDT session or connecting to an existing AEDT session, this API provides methods for automating the segmentation and skew of a 3D motor.

Warning

Both segmentation and skew of a 3D motor have requirements on the AEDT active project. Ensure that the active design meets these requirements:

- For segmentation, `SymmetryFactor` and `HalfAxial` design settings must be defined.
- For skew, `Shaft` must be the name of the shaft.

`ToolkitBackend()`

Provides methods for controlling the toolkit workflow.

3.1.1 `ansys.aedt.toolkits.magnet_segmentation.backend.api.ToolkitBackend`

class `ansys.aedt.toolkits.magnet_segmentation.backend.api.ToolkitBackend`

Provides methods for controlling the toolkit workflow.

This class provides methods for creating an AEDT session, connecting to an existing AEDT session, and automating the segmentation and skew of a 3D motor model.

Examples

```
>>> from ansys.aedt.toolkits.magnet_segmentation.backend.api import ToolkitBackend
>>> toolkit = ToolkitBackend()
>>> toolkit.launch_aedt()
```

`__init__()`

Methods

<code>__init__()</code>	
<code>aedt_sessions()</code>	Get information for the active AEDT sessions.
<code>apply_skew()</code>	Apply skew to rotor slices.
<code>connect_aedt()</code>	Connect to an existing AEDT session.
<code>connect_design([app_name])</code>	Connect to an application design.
<code>export_aedt_model([obj_list, export_path, ...])</code>	Export the model in the OBJ format and then encode the file if the encode parameter is enabled.
<code>get_design_names()</code>	Get the design names for a specific project.
<code>get_magnet_loss()</code>	Get magnet loss.
<code>get_project_name(project_path)</code>	Get the project name from the project path.
<code>get_properties()</code>	Get the toolkit properties.
<code>get_thread_status()</code>	Get the toolkit thread status.
<code>installed_aedt_version()</code>	Get the installed AEDT versions.
<code>is_aedt_connected()</code>	Check if AEDT is connected.
<code>launch_aedt()</code>	Launch AEDT.
<code>launch_thread(process, *args)</code>	Launch the thread.
<code>open_project([project_name])</code>	Open an AEDT project.
<code>release_aedt([close_projects, close_on_exit])</code>	Release AEDT.
<code>save_project([project_path, release_aedt])</code>	Save the project.
<code>segmentation()</code>	Apply object segmentation.
<code>serialize_obj_base64(file_path)</code>	Encode a bytes-like object.
<code>set_properties(data)</code>	Assign the passed data to the internal data model.
<code>validate_and_analyze()</code>	Validate and analyze the design.
<code>wait_to_be_idle([timeout])</code>	Wait for the thread to be idle and ready to accept a new task.

This code shows how to use the ToolkitBackend class:

```
# Import required modules and backend
from ansys.aedt.toolkits.magnet_segmentation.backend.api import ToolkitBackend

# Initialize generic service
toolkit = ToolkitBackend()

# Get the default properties loaded from JSON file
properties = toolkit.get_properties()

# Set properties
aedt_file = os.path.join(common_temp_dir, "input_data", f"{PROJECT_NAME}.aedt")
new_properties = {
    "aedt_version": "2024.1",
```

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```
"non_graphical": True,
"active_project": aedt_file,
"active_design": DESIGN_NAME,
"design_list": {PROJECT_NAME: [DESIGN_NAME]},
"use_grpc": config["use_grpc"],
}
toolkit.set_properties(new_properties)

# Launch AEDT and open project
msg = toolkit.launch_aedt()
toolkit.open_project(aedt_file)
toolkit.connect_design("Maxwell3D")

# Segment and skew motor
toolkit.segmentation()
toolkit.apply_skew()

# Release AEDT
toolkit.release_aedt()
```


EXAMPLES

End-to-end example show how you can use the Magnet Segmentation Toolkit.

4.1 Maxwell 3D segmentation

This example shows how to use the Magnet Segmentation Toolkit to segment your AEDT motor model.

4.1.1 Perform required imports

Perform required imports.

```
[1]: import shutil
import tempfile
from pathlib import Path

from ansys.aedt.toolkits.common.utils import download_file

from ansys.aedt.toolkits.magnet_segmentation.backend.api import ToolkitBackend
```

4.1.2 Initialize temporary folder and project settings

Initialize a temporary folder to copy the input file into and specify project settings.

```
[2]: URL_BASE = "https://raw.githubusercontent.com/ansys/example-data/master/toolkits/magnet_
      ↪segmentation/"
AEDT_PROJECT = "e9_eMobility_IPM_3D"
URL = str(Path(URL_BASE) / (AEDT_PROJECT + ".aedt"))

temp_dir = tempfile.TemporaryDirectory(suffix=".ansys")
active_project = str(Path(temp_dir.name) / (AEDT_PROJECT + ".aedt"))
download_file(URL, active_project)
active_design = "e9_eMobility_IPM_3D_test"
```

```
-----
InvalidURL                                Traceback (most recent call last)
Cell In[2], line 7
      5 temp_dir = tempfile.TemporaryDirectory(suffix=".ansys")
      6 active_project = str(Path(temp_dir.name) / (AEDT_PROJECT + ".aedt"))
----> 7 download_file(URL, active_project)
```

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```

8 active_design = "e9_eMobility_IPM_3D_test"

File C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
↳venv\Lib\site-packages\ansys\aedt\toolkits\common\utils.py:45, in download_file(url,
↳local_filename)
    43 """Download a file from a URL into a local file."""
    44 download_timeout = DEFAULT_REQUESTS_TIMEOUT * 2
--> 45 with requests.get(url, stream=True, timeout=download_timeout) as r:
    46     r.raise_for_status()
    47     with open(local_filename, "wb") as f:

File C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
↳venv\Lib\site-packages\requests\api.py:87, in get(url, params, **kwargs)
    74 def get(
    75     url: _t.UriType, params: _t.ParamsType = None, **kwargs: Unpack[_t.GetKwargs]
    76 ) -> Response:
    77     r"""Sends a GET request.
    78
    79     :param url: URL for the new :class:`Request` object.
    (... ) 84     :rtype: requests.Response
    85     """
--> 87     return request(, url, params=params, **kwargs)

File C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
↳venv\Lib\site-packages\requests\api.py:71, in request(method, url, **kwargs)
    67 # By using the 'with' statement we are sure the session is closed, thus we
    68 # avoid leaving sockets open which can trigger a ResourceWarning in some
    69 # cases, and look like a memory leak in others.
    70 with sessions.Session() as session:
--> 71     return session.request(method=method, url=url, **kwargs)

File C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
↳venv\Lib\site-packages\requests\sessions.py:635, in Session.request(self, method, url,
↳params, data, headers, cookies, files, auth, timeout, allow_redirects, proxies, hooks,
↳stream, verify, cert, json)
    622 # Create the Request.
    623 req = Request(
    624     method=method.upper(),
    625     url=url,
    (... ) 633     hooks=hooks,
    634 )
--> 635 prep = self.prepare_request(req)
    637 assert _is_prepared(prepare)
    639 proxies = proxies or {}

File C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
↳venv\Lib\site-packages\requests\sessions.py:541, in Session.prepare_request(self,
↳request)
    538     auth = get_netrc_auth(url)
    540 p = PreparedRequest()
--> 541 p.prepare(
    542     method=method.upper(),

```

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```

543     url=url,
544     files=request.files,
545     data=request.data,
546     json=request.json,
547     headers=merge_setting(
548         request.headers, self.headers, dict_class=CaseInsensitiveDict
549     ),
550     params=merge_setting(request.params, self.params),
551     auth=merge_setting(auth, self.auth),
552     cookies=merged_cookies,
553     hooks=merge_hooks(request.hooks, self.hooks),
554 )
555 return p

```

File C:\actions-runner_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\.
 ↳venv\Lib\site-packages\requests\models.py:439, in PreparedRequest.prepare(self, method,
 ↳url, headers, files, data, params, auth, cookies, hooks, json)

```

437 url = cast("_t.UriType", url)
438 self.prepare_method(method)
--> 439 self.prepare_url(url, params)
440 self.prepare_headers(headers)
441 self.prepare_cookies(cookies)

```

File C:\actions-runner_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\.
 ↳venv\Lib\site-packages\requests\models.py:520, in PreparedRequest.prepare_url(self,
 ↳url, params)

```

514     raise MissingSchema(
515         f"Invalid URL {url!r}: No scheme supplied. "
516         f"Perhaps you meant https://{url}?"
517     )
519 if not host:
--> 520     raise InvalidURL(f"Invalid URL {url!r}: No host supplied")
522 # In general, we want to try IDNA encoding the hostname if the string contains
523 # non-ASCII characters. This allows users to automatically get the correct IDNA
524 # behaviour. For strings containing only ASCII characters, we need to also verify
525 # it doesn't start with a wildcard (*), before allowing the unencoded hostname.
526 if not unicode_is_ascii(host):

```

InvalidURL: Invalid URL 'https:\\raw.githubusercontent.com\\ansys\\example-data\\master\\
 ↳toolkits\\magnet_segmentation\\e9_eMobility_IPM_3D.aedt': No host supplied

4.1.3 Initialize toolkit

Initialize the toolkit.

```
[3]: toolkit = ToolkitBackend()
```

4.1.4 Get toolkit properties

Get the toolkit properties.

```
[4]: properties = toolkit.get_properties()
```

4.1.5 Initialize properties

Initialize a dictionary of properties.

```
[5]: properties["aedt_version"] = "2026.1"
properties["active_project"] = AEDT_PROJECT
properties["active_design"] = active_design
properties["design_list"] = {AEDT_PROJECT: [active_design]}
properties["is_skewed"] = False
properties["rotor_material"] = "M250-35A_20C"
properties["stator_material"] = "M250-35A_20C"
properties["magnets_material"] = "N30UH_65C"
properties["magnet_segments_per_slice"] = 2
properties["rotor_slices"] = 2
properties["apply_mesh_sheets"] = False
# properties["mesh_sheets_number"] = 3
properties["skew_angle"] = "2deg"
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[5], line 3
      1 properties["aedt_version"] = "2026.1"
      2 properties["active_project"] = AEDT_PROJECT
----> 3 properties["active_design"] = active_design
      4 properties["design_list"] = {AEDT_PROJECT: [active_design]}
      5 properties["is_skewed"] = False

NameError: name 'active_design' is not defined
```

4.1.6 Set non-graphical mode

Set non-graphical mode. The default value is False.

```
[6]: properties["non_graphical"] = False
```

4.1.7 Set properties

Set properties.

```
[7]: toolkit.set_properties(properties)
INFO - Updating internal properties.
```

```
[7]: (True, 'Properties were updated successfully.')
```

4.1.8 Initialize AEDT

Launch a new AEDT session.

```
[8]: toolkit.launch_aedt()
INFO - AEDT is released.
[8]: True
```

4.1.9 Open project

Open the project.

```
[9]: toolkit.open_project(active_project)

-----
AttributeError                                Traceback (most recent call last)
File C:\actions-runner\work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
venv\Lib\site-packages\ansys\aedt\core\internal\grpc_plugin_dll_class.py:116, in
AedtObjWrapper.__Invoke__(self, funcName, argv)
    115     self.dllapi.recreate_application(True)
--> 116 ret = _retry_ntimes(
    117     settings.number_of_grpc_api_retries,
    118     self.dllapi.AedtAPI.InvokeAedtObjMethod,
    119     self.objectID,
    120     funcName,
    121     argv,
    122 ) # Call C function
    123 if ret and isinstance(ret, (AedtObjWrapper, AedtPropServer)):

File C:\actions-runner\work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
venv\Lib\site-packages\ansys\aedt\core\generic\general_methods.py:613, in _retry_
ntimes(n, function, *args, **kwargs)
    612 if "__name__" in dir(function):
--> 613     raise AttributeError(f"Error in Executing Method {function.__name__}.")
    614 else:
```

AttributeError: Error in Executing Method InvokeAedtObjMethod.

During handling of the above exception, another exception occurred:

```
GrpcApiError                                Traceback (most recent call last)
Cell In[9], line 1
----> 1 toolkit.open_project(active_project)

File C:\actions-runner\work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
venv\Lib\site-packages\ansys\aedt\toolkits\common\backend\api.py:743, in AEDTCommon.
open_project(self, project_name)
    741     return True
    742 elif not os.path.exists(project_name + ".lock") and self.desktop and project_
name:
--> 743     self.desktop.odesktop.OpenProject(project_name)
    744     logger.debug("Project {} is opened".format(project_name))
    745     self.__save_project_info()
```

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```

File C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
→venv\Lib\site-packages\ansys\aedt\core\internal\grpc_plugin_dll_class.py:138, in_
→AedtObjWrapper.__GetObjMethod__.<locals>.DynamicFunc(self, *args)
    137 def DynamicFunc(self, *args):
--> 138     return self.__Invoke__(funcName, args)

File C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
→venv\Lib\site-packages\ansys\aedt\core\internal\grpc_plugin_dll_class.py:129, in_
→AedtObjWrapper.__Invoke__(self, funcName, argv)
    127 pyaedt_logger.debug("Failed to execute gRPC AEDT command:")
    128 pyaedt_logger.debug(f"{funcName}({argv})")
--> 129 raise GrpcApiError(f"Failed to execute gRPC AEDT command: {funcName}")

GrpcApiError: Failed to execute gRPC AEDT command: OpenProject

```

4.1.10 Connect design

Connect or create a new design.

```

[10]: toolkit.connect_design()
INFO - Updating internal properties.
INFO - Toolkit is connected to AEDT design.

[10]: True

```

4.1.11 Apply segmentation

Apply segmentation and assign the relative coordinate system.

```

[11]: toolkit.segmentation()
INFO - AEDT is released.
INFO - Toolkit is connected to AEDT design.

PyAEDT ERROR: *****
PyAEDT ERROR:   File "<frozen runpy>", line 198, in _run_module_as_main
PyAEDT ERROR:   File "<frozen runpy>", line 88, in _run_code
PyAEDT ERROR:   File "C:\actions-runner\_work\_tool\Python\3.12.10\X64\Lib\asyncio\base_
→events.py", line 645, in run_forever
PyAEDT ERROR:       self._run_once()
PyAEDT ERROR:   File "C:\actions-runner\_work\_tool\Python\3.12.10\X64\Lib\asyncio\base_
→events.py", line 1999, in _run_once
PyAEDT ERROR:       handle._run()
PyAEDT ERROR:   File "C:\actions-runner\_work\_tool\Python\3.12.10\X64\Lib\asyncio\
→events.py", line 88, in _run
PyAEDT ERROR:       self._context.run(self._callback, *self._args)
PyAEDT ERROR:   File "C:\Users\ansys\AppData\Local\Temp\ipykernel_11872\450086471.py", l_
→ine 1, in <module>
PyAEDT ERROR:       toolkit.segmentation()
PyAEDT ERROR:   File "C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-
→segmentation-toolkit\venv\Lib\site-packages\ansys\aedt\toolkits\magnet_segmentation\
→backend\workflows\aedt.py", line 74, in segmentation

```

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```

PyAEDT ERROR:      if self.aedtapp["HalfAxial"] == "1":
PyAEDT ERROR:      File "C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-
↳segmentation-toolkit\.venv\Lib\site-packages\ansys\aedt\core\application\design.py",
↳line 329, in __getitem__
PyAEDT ERROR:      return self.variable_manager[variable_name].expression
PyAEDT ERROR:      File "C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-
↳segmentation-toolkit\.venv\Lib\site-packages\ansys\aedt\core\application\variables.py",
↳line 768, in __getitem__
PyAEDT ERROR:      return self.variables[variable_name]
PyAEDT ERROR:      ~~~~~~^~~~~~
PyAEDT ERROR: 'halfaxial' on __getitem__
PyAEDT ERROR: Method arguments:
PyAEDT ERROR:      variable_name = HalfAxial
PyAEDT ERROR: *****
PyAEDT ERROR: *****
PyAEDT ERROR:      File "<frozen runpy>", line 198, in _run_module_as_main
PyAEDT ERROR:      File "<frozen runpy>", line 88, in _run_code
PyAEDT ERROR:      File "C:\actions-runner\_work\_tool\Python\3.12.10\x64\Lib\asyncio\base_
↳events.py", line 645, in run_forever
PyAEDT ERROR:      self._run_once()
PyAEDT ERROR:      File "C:\actions-runner\_work\_tool\Python\3.12.10\x64\Lib\asyncio\base_
↳events.py", line 1999, in _run_once
PyAEDT ERROR:      handle._run()
PyAEDT ERROR:      File "C:\actions-runner\_work\_tool\Python\3.12.10\x64\Lib\asyncio\
↳events.py", line 88, in _run
PyAEDT ERROR:      self._context.run(self._callback, *self._args)
PyAEDT ERROR:      File "C:\Users\ansys\AppData\Local\Temp\ipykernel_11872\450086471.py",
↳line 1, in <module>
PyAEDT ERROR:      toolkit.segmentation()
PyAEDT ERROR:      File "C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-
↳segmentation-toolkit\.venv\Lib\site-packages\ansys\aedt\toolkits\magnet_segmentation\
↳backend\workflows\aedt.py", line 74, in segmentation
PyAEDT ERROR:      if self.aedtapp["HalfAxial"] == "1":
PyAEDT ERROR:      File "C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-
↳segmentation-toolkit\.venv\Lib\site-packages\ansys\aedt\core\application\design.py",
↳line 329, in __getitem__
PyAEDT ERROR:      return self.variable_manager[variable_name].expression
PyAEDT ERROR:      ~~~~~~^~~~~~
PyAEDT ERROR:      File "C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-
↳segmentation-toolkit\.venv\Lib\site-packages\ansys\aedt\core\application\variables.py",
↳line 768, in __getitem__
PyAEDT ERROR:      return self.variables[variable_name]
PyAEDT ERROR:      ~~~~~~^~~~~~
PyAEDT ERROR: 'halfaxial' on __getitem__
PyAEDT ERROR: Method arguments:
PyAEDT ERROR:      variable_name = HalfAxial
PyAEDT ERROR: *****

```

```

-----
KeyError                                Traceback (most recent call last)
Cell In[11], line 1
----> 1 toolkit.segmentation()

```

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```

File C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
→venv\Lib\site-packages\ansys\aedt\toolkits\magnet_segmentation\backend\workflows\aedt.
→py:74, in AEDTWorkflow.segmentation(self)
    71 self.connect_design()
    73 # Requirements: Design needs a design variable "HalfAxial"
--> 74 if self.aedtapp[ ] == "1":
    75     self.aedtapp["HalfAxial"] = "0"
    77 for obj in self.aedtapp.modeler.unclassified_objects:

```

```

File C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
→venv\Lib\site-packages\ansys\aedt\core\generic\general_methods.py:252, in _function_
→handler_wrapper.<locals>.wrapper(*args, **kwargs)
    250 msg = msg.capitalize()
    251 _exception(sys.exc_info(), user_function, args, kwargs, msg)
--> 252 return raise_exception_or_return_false(e)

```

```

File C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
→venv\Lib\site-packages\ansys\aedt\core\generic\general_methods.py:214, in raise_
→exception_or_return_false(e)
    211 for v in list(_desktop_sessions.values())[ : ]:
    212     v.release_desktop(close_projects=v.close_on_exit, close_on_exit=v.
→close_on_exit)
--> 214 raise e
    215 elif "__init__" in str(e): # pragma: no cover
    216     return

```

```

File C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
→venv\Lib\site-packages\ansys\aedt\core\generic\general_methods.py:227, in _function_
→handler_wrapper.<locals>.wrapper(*args, **kwargs)
    225 try:
    226     settings.time_tick = time.time()
--> 227 out = user_function(*args, **kwargs)
    228 _log_method(user_function, args, kwargs)
    229 return out

```

```

File C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
→venv\Lib\site-packages\ansys\aedt\core\application\design.py:329, in Design.__getitem__
→(self, variable_name)
    327 @pyaedt_function_handler()
    328 def __getitem__(self, variable_name: str) -> str:
--> 329 return self.variable_manager[variable_name].expression

```

```

File C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
→venv\Lib\site-packages\ansys\aedt\core\generic\general_methods.py:252, in _function_
→handler_wrapper.<locals>.wrapper(*args, **kwargs)
    250 msg = msg.capitalize()
    251 _exception(sys.exc_info(), user_function, args, kwargs, msg)
--> 252 return raise_exception_or_return_false(e)

```

```

File C:\actions-runner\_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
→venv\Lib\site-packages\ansys\aedt\core\generic\general_methods.py:214, in raise_

```

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```

↪exception_or_return_false(e)
    211         for v in list(_desktop_sessions.values())[::]:
    212             v.release_desktop(close_projects=v.close_on_exit, close_on_exit=v.
↪close_on_exit)
--> 214         raise e
    215     elif "__init__" in str(e): # pragma: no cover
    216         return

```

File C:\actions-runner_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
↪venv\Lib\site-packages\ansys\aedt\core\generic\general_methods.py:227, in _function_
↪handler_wrapper.<locals>.wrapper(*args, **kwargs)

```

    225 try:
    226     settings.time_tick = time.time()
--> 227     out = user_function(*args, **kwargs)
    228     _log_method(user_function, args, kwargs)
    229     return out

```

File C:\actions-runner_work\magnet-segmentation-toolkit\magnet-segmentation-toolkit\
↪venv\Lib\site-packages\ansys\aedt\core\application\variables.py:768, in_
↪VariableManager.__getitem__(self, variable_name)

```

    766 @pyaedt_function_handler()
    767 def __getitem__(self, variable_name):
--> 768     return self.variables[variable_name]

```

KeyError: 'HalfAxial'

4.1.12 Apply skew angle

Apply the skew angle to rotor slices.

```
[12]: toolkit.apply_skew()
```

```
INFO - AEDT is released.
```

```
[12]: False
```

4.1.13 Validate and analyze design

Uncomment the line to validate and analyze the design.

```
toolkit.validate_and_analyze()
```

4.1.14 Create magnet loss report

Uncomment the lines to create magnet loss report and compute average value.

```
magnet_loss = toolkit.get_magnet_loss() print(fAverage magnet loss: {magnet_loss} W)
```

4.1.15 Save and release AEDT

Save and release AEDT.

```
toolkit.save_project()
```

```
[13]: toolkit.release_aedt(True, True)
```

```
INFO - AEDT is released.
```

```
[13]: True
```

4.1.16 Remove temporary folder

Remove the temporary folder.

```
[14]: shutil.rmtree(temp_dir.name, ignore_errors=True)
```

CONTRIBUTE

Overall guidance on contributing to a PyAnsys repository appears in [Contributing](#) in the *PyAnsys developers guide*. Ensure that you are thoroughly familiar with this guide before attempting to contribute to PyAEDT or its toolkits.

The following contribution information is specific to the Magnet Segmentation Toolkit.

5.1 Clone the repository

To clone and install the latest version of the Magnet Segmentation Toolkit in development mode, run these commands:

```
git clone https://github.com/ansys/magnet-segmentation-toolkit.git
cd magnet-segmentation-toolkit
python -m pip install --upgrade pip
pip install -e .
```

5.2 Post issues

Use the [Magnet Segmentation Toolkit Issues](#) page to create issues to report bugs and request new features.

5.3 View documentation

Documentation for the latest stable release is hosted at [Magnet Segmentation Toolkit Documentation](#).

In the upper right corner of the documentations title bar, there is an option for switching from viewing the documentation for the latest stable release to viewing the documentation for the development version or previously released versions.

5.4 Adhere to code style

The Magnet Segmentation Toolkit is compliant with [PyAnsys code style](#). It uses the tool [pre-commit](#) to check the code style.

You can install and activate this tool with these commands:

```
pip install pre-commit
pre-commit run --all-files
```

You can also install this as a pre-commit hook with this command:

```
pre-commit install
```

This way, its not possible for you to push code that fails the style checks:

```
$ pre-commit install
$ git commit -am "Add my cool feature."
black.....Passed
isort (python).....Passed
flake8.....Passed
codespell.....Passed
fix requirements.txt.....Passed
blacken-docs.....Passed
```

5.4.1 Maximum line length

Best practice is to keep the line length at or below 120 characters for code and comments. Lines longer than this might not display properly on some terminals and tools or might be difficult to follow.

RELEASE NOTES

This document contains the release notes for the project.

6.1 0.9.0 - April 23, 2026

Documentation

Fix vtk-osmesa install step	#283
-----------------------------	------

Maintenance

Update changelog for v0.8.5	#276
Update missing or outdated files	#286
Pin versions of actions in GitHub workflow files with SHA	#288
Add dependabot cooldown settings	#296
Updates	#297
Bump actions versions	#303
Upload new ci_cd yml files	#305

6.2 0.8.5 - July 17, 2025

Fixed

Fix download links	#274
--------------------	------

Maintenance

Update changelog for v0.8.3	#273
-----------------------------	------

6.3 0.8.4 - July 17, 2025

No significant changes. 0.8.3 - July 10, 2025 =====

Dependencies

Bump actions/download-artifact from 4.1.9 to 4.3.0	#259
--	------

Fixed

Ci cd	#271
-------	------

Maintenance

Update changelog for v0.8.2	#269
-----------------------------	------

6.4 0.8.2 - July 08, 2025

Fixed

Populate props	#268
----------------	------

Maintenance

Update changelog for v0.8.1	#265
-----------------------------	------

6.5 0.8.1 - June 13, 2025

Dependencies

Bump ansys/actions from 9 to 10	#262
---------------------------------	------

Documentation

update installation.rst	#263
-------------------------	------

Maintenance

update CHANGELOG for v0.8.0	#260
update - 0.9.dev0	#261

6.6 0.8.0 - June 03, 2025

Fixed

improve skew	#251
rotate insulation sheet + update cicc pypi	#258

Maintenance

update CHANGELOG for v0.7.0	#249
Update v0.8.dev0	#250
update cicc and add linux tests	#254
add codecov.yml	#256

6.7 0.7.0 - May 08, 2025**Documentation**

Update CONTRIBUTORS.md with the latest contributors	#243
---	------

Fixed

Fix release artifacts	#248
-----------------------	------

Miscellaneous

Create installer	#245
------------------	------

INDICES AND TABLES

- `genindex`
- `modindex`
- `search`